Q1. Define Image Compression. Why is Image Compression Required in Digital Image Processing?

Definition

Image compression is the process of reducing the size of a digital image by reducing the number of bits needed to store or transmit it.

Why is Image Compression Required?

1. Reduces Storage Space
   It reduces the size of image files, so more images can be stored.
2. Faster Transmission
   Smaller image files can be transferred faster over the internet.
3. Saves Bandwidth
   Compressed images require less bandwidth during transmission.
4. Reduces Processing Time
   Smaller files take less time to transfer and process.
5. Maintains Image Quality
   Compression can reduce file size while maintaining suitable image quality.

Types of Image Compression

1. Lossless Compression

- No information is lost.
- Original image can be recovered exactly.
- Example: PNG

2. Lossy Compression

- Some image information is removed.
- Gives a smaller file size.
- Example: JPEG

Example

A 5 MB image can be compressed to around 1 MB, making it easier to store and transmit.

Q2. Explain the Different Types of Redundancies in Digital Images with Suitable Examples.

Redundancy means repeated or unnecessary information present in an image. Removing redundancy helps to reduce image size.

1. Coding Redundancy

Coding redundancy occurs when more bits than necessary are used to represent image information.

Example:
Frequently occurring symbols can be given shorter codes using Huffman Coding.

For example:

A → 0
B → 10

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2. Inter-Pixel Redundancy

Inter-pixel redundancy occurs because neighboring pixels are usually similar or related.

For example, pixels in a clear sky may have almost the same intensity.

Example:

A A A A B B B

Using RLE:

A4 B3

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3. Psycho-Visual Redundancy

Psycho-visual redundancy occurs because the human eye cannot notice every small detail in an image.

Less important visual information can be reduced during compression.

Example:
JPEG compression removes some less noticeable image information.

Main Types

- Coding redundancy → Extra coding information
- Inter-pixel redundancy → Similar neighboring pixels
- Psycho-visual redundancy → Less noticeable visual information

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Q3. Explain the Working Principle of Run-Length Encoding (RLE) with a Suitable Example.

Definition

Run-Length Encoding (RLE) is a lossless image compression technique that stores a repeated pixel value along with the number of times it occurs continuously.

Working Principle

1. Read the pixels sequentially.
2. Find continuously repeated pixels.
3. Count the repeated pixels.
4. Store the pixel value and its count.
5. Continue until all pixels are processed.

Example

Suppose the image contains:

A A A A A B B B C C

Using RLE:

A5 B3 C2

Here:

- A5 → A is repeated 5 times
- B3 → B is repeated 3 times
- C2 → C is repeated 2 times

Grayscale Example

Original:

10 10 10 20 20 30 30 30

RLE:

(10,3) (20,2) (30,3)

Advantages

- Simple and easy to implement.
- It is a lossless method.
- Encoding and decoding are fast.
- Works well for images with large repeated areas.

Limitation

RLE is not effective for images where pixel values change frequently.

Q4. Apply Huffman Coding to a Given Set of Image Symbols and Demonstrate How Variable-Length Codes Reduce the Number of Bits Required for Storage.

Definition

Huffman Coding is a lossless compression technique that assigns shorter codes to frequently occurring symbols and longer codes to less frequent symbols.

Example

Consider:

Symbol	Frequency
A	50
B	25
C	15
D	10

One possible Huffman code is:

Symbol	Code
A	0
B	10
C	110
D	111

The most frequent symbol A gets the shortest code.

Calculation

- A → 50 × 1 = 50 bits

- B → 25 × 2 = 50 bits
- C → 15 × 3 = 45 bits
- D → 10 × 3 = 30 bits

Total = 175 bits

With fixed-length coding, 4 symbols require 2 bits per symbol:

100 × 2 = 200 bits

Therefore:

Fixed-length = 200 bits
Huffman Coding = 175 bits

So, Huffman Coding saves 25 bits.

Main Point

Huffman Coding reduces storage by using variable-length codes, where frequently occurring symbols get shorter codes.

Q5. Compare RLE, Huffman Coding, and LZW in Terms of Working Principle, Compression Efficiency, Advantages, and Limitations.

Feature	RLE	Huffman Coding	LZW
Principle	Stores repeated values with count	Uses different code lengths	Uses a dictionary of repeated patterns
Type	Lossless	Lossless	Lossless
Efficiency	Good for repeated data	Generally good	Good for repeated patterns
Advantage	Simple and fast	Efficient compression	Compresses repeated sequences
Limitation	Poor for non-repetitive data	Needs frequency calculation	Needs dictionary management

RLE

RLE represents repeated values using their count.

Example:
AAAAAA → A6

It is best for images having large areas of the same pixel value.

Huffman Coding

Huffman Coding gives short codes to frequent symbols and longer codes to less frequent symbols.

Example:
A → 0, B → 10, C → 110

LZW

LZW (Lempel-Ziv-Welch) creates a dictionary of repeated patterns and replaces repeated sequences with dictionary codes.

Key Difference

- RLE → Repeated pixels
- Huffman → Symbol frequency
- LZW → Repeated patterns/dictionary

Q6. Explain the Basic Steps Involved in JPEG Image Compression, Including DCT, Quantization, and Entropy Coding.

JPEG (Joint Photographic Experts Group) is a commonly used **lossy image compression** method. It reduces image size while maintaining good visual quality.

Steps of JPEG Compression

1. Divide Image into Blocks

- The image is converted into suitable color format.
- It is divided into small **8 × 8 pixel blocks**.

2. DCT (Discrete Cosine Transform)

- DCT converts the image data from the **spatial domain to frequency domain**.
- It separates important image information into different frequency components.
- Low-frequency components contain major image details.

3. Quantization

- DCT values are divided by **quantization values**.
- Less important high-frequency information is reduced or removed.
- This is the main **lossy step** in JPEG compression.

4. Zig-Zag Scanning

- The quantized values are arranged in a **zig-zag order**.
- This places low-frequency values before high-frequency values.

5. Entropy Coding

- The data is further compressed using techniques such as **Huffman Coding**.
- Frequently occurring values get shorter codes.

Simple Flow

Image → 8×8 Blocks → DCT → Quantization → Zig-Zag Scanning → Entropy Coding → Compressed Image

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Q7. Analyze the Difference Between Lossless and Lossy Image Compression. Discuss Suitable Applications for Each Type.

Image compression is mainly divided into **lossless** and **lossy** compression.

1. Lossless Compression

In lossless compression, **no image information is permanently lost**. The original image can be recovered exactly after decompression.

Examples:

- PNG
- RLE
- Huffman Coding
- LZW

Advantages:

- Original image is recovered perfectly.
- No loss of important information.
- Suitable when accuracy is important.

Applications:

- Medical images
- Technical drawings
- Documents
- Scientific images

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2. Lossy Compression

In lossy compression, **some image information is removed** to achieve a much smaller file size.

Example:

- JPEG

Advantages:

- Provides high compression.
- Requires less storage space.
- Faster image transmission.

Applications:

- Web images
- Digital photographs
- Social media
- Multimedia applications

Difference

Feature	Lossless	Lossy
Data loss	No	Yes
Original image	Fully recovered	Not exactly recovered
File size	Larger	Smaller
Quality	Same	Slightly reduced

Feature	Lossless	Lossy
Example	PNG	JPEG
Use	Medical/documents	Photos/web images

Q8. Define Image Segmentation. List the Major Techniques Used for Segmenting Digital Images.

Definition

Image segmentation is the process of **dividing an image into meaningful regions or objects** so that each region contains similar or related information.

The main purpose is to separate the **objects of interest from the background**.

Major Techniques

1. Thresholding

- Divides an image based on pixel intensity.
- Commonly used for separating objects from the background.

2. Edge-Based Segmentation

- Detects boundaries or edges of objects.
- Operators such as **Sobel and Canny** can be used.

3. Region-Based Segmentation

- Groups neighboring pixels having similar properties.
- Example: **Region Growing**.

4. Clustering-Based Segmentation

- Groups pixels into clusters based on similar characteristics such as intensity or color.

5. Watershed Segmentation

- Treats the image like a surface and separates different regions.

6. Hough Transform

- Used to detect specific shapes such as **lines and circles**.

Example

In a medical image, segmentation can be used to separate a **tumor from surrounding tissues.**

Q9. Explain the Working Principle of the Sobel Edge Detection Operator. What Type of Information Does It Extract from an Image?

Definition

The **Sobel operator** is an edge detection technique used to find **edges and boundaries of objects** in an image.

It detects changes in pixel intensity in the **horizontal and vertical directions**.

Working Principle

1. Convert to Grayscale

- The input image is generally converted into a grayscale image.

2. Apply Sobel Masks
Two 3×3 masks are used:

Horizontal Mask (Gx):

-1 0 1

-2 0 2

-1 0 1

Vertical Mask (Gy):

-1 -2 -1

0 0 0

1 2 1

3. Calculate Gradients

- Gx detects changes in the **horizontal direction**.
- Gy detects changes in the **vertical direction.**

4. Find Edge Strength
The two gradients are combined to determine the strength of an edge.

Information Extracted

Sobel mainly extracts:

- **Edges**
- **Object boundaries**
- **Intensity changes**
- Horizontal and vertical edge information

Example

If there is a sharp change from a dark region to a bright region, the Sobel operator produces a strong edge at that location.

Q10. Explain the Major Steps Involved in the Canny Edge Detection Algorithm.

Definition

Canny Edge Detection is an edge detection method used to detect **clear and accurate edges** in an image.

Major Steps

1. Noise Reduction

- The image is first smoothed using a **Gaussian filter**.
- This reduces noise and unwanted details.

2. Gradient Calculation

- The intensity changes in the image are calculated.
- This helps identify possible edges.

3. Non-Maximum Suppression

- Weak or unnecessary pixels are removed.
- Only the strongest pixels along an edge are kept.

4. Double Thresholding
Two threshold values are used:

- **High threshold** → strong edges
- **Low threshold** → weak edges

5. Edge Tracking by Hysteresis

- Weak edges connected to strong edges are kept.
- Weak edges not connected to strong edges are removed.

Simple Flow

Input Image → Gaussian Filtering → Gradient Calculation → Non-Maximum Suppression → Double Thresholding → Edge Tracking → Final Edges

Main Advantage

Canny produces **thin, clear, and well-localized edges** and is less affected by noise compared with simpler edge detection methods.

Q11. Compare Sobel and Canny Edge Detection with Respect to Noise Sensitivity, Edge Localization, Computational Complexity, and Output Quality.

Sobel Edge Detection

- Uses **horizontal and vertical masks** to detect edges.
- It is **more sensitive to noise.**
- Edge localization is **less accurate.**
- It is **simple and fast** to calculate.
- Output may contain **thicker or less clear edges.**

Canny Edge Detection

- Uses several steps such as **Gaussian filtering, gradient calculation, non-maximum suppression, and thresholding.**
- It is **less sensitive to noise** because smoothing is performed first.
- Provides **better edge localization.**
- It is **more computationally complex** than Sobel.
- Produces **thin, clear, and accurate edges.**

Comparison

Feature	Sobel	Canny
Noise sensitivity	More	Less
Edge localization	Less accurate	More accurate

Feature	Sobel	Canny
Complexity	Low	Higher
Output quality	Good	Very good
Processing speed	Faster	Slower

Q12. Explain How Global Thresholding Can Be Applied to Separate an Object from Its Background in a Grayscale Image. Discuss the Factors Affecting the Selection of Threshold Value.

Definition

Global thresholding is a segmentation technique that uses a **single threshold value** to separate an object from its background in a grayscale image.

Working

1. Select a threshold value **T**.
2. Check the intensity of each pixel.
3. If the pixel value is greater than or equal to T, assign it to the object.
4. If the pixel value is less than T, assign it to the background.

The basic rule is:

- **Pixel ≥ T → Object**
- **Pixel < T → Background**

Example

Suppose the threshold is **T = 100**.

- Pixel value **150 → Object**
- Pixel value **80 → Background**
- Pixel value **120 → Object**
- Pixel value **50 → Background**

Factors Affecting Threshold Selection

1. **Object and Background Intensity** – A clear difference makes threshold selection easier.
2. **Image Contrast** – Higher contrast generally gives better segmentation.
3. **Lighting Conditions** – Uneven illumination can make a single threshold unsuitable.
4. **Noise** – Noise can affect pixel intensity and produce incorrect segmentation.
5. **Image Histogram** – Histogram distribution helps in selecting a suitable threshold.

Application

Global thresholding is useful when the **object and background have clearly different intensity values**.

Q13. Describe How Otsu's Thresholding Method Can Be Used to Automatically Determine a Suitable Threshold for Image Segmentation.

Definition

Otsu's thresholding is an automatic thresholding method used to separate an image into **object and background**.

It automatically selects the best threshold value from the image histogram.

Working Steps

1. Convert the image into a **grayscale image**.
2. Calculate the **histogram** of the image.
3. Consider different possible threshold values.
4. Divide the pixels into two classes:
    - **Object**
    - **Background**
5. Calculate the variance between these two classes.
6. Select the threshold that gives the **maximum between-class variance**.
7. Use this threshold to create the segmented image.

Example

Suppose an image contains a dark object on a bright background.

Otsu's method analyzes the intensity distribution and automatically selects a suitable threshold, for example **T = 120**.

Then:

- Pixel ≥ 120 → Background
- Pixel < 120 → Object

Advantages

- Threshold is selected **automatically**.
- No need to manually choose the threshold.
- Works well when object and background have **different intensity distributions**.
- Simple and widely used for image segmentation.

Q14. An Image Contains Objects Under Non-Uniform Illumination. Analyze the Suitability of Global, Otsu's, and Adaptive Thresholding Methods and Justify Which Method Would Provide Better Segmentation.

When an image has **non-uniform illumination**, different parts of the image may have different brightness levels. Therefore, choosing one threshold for the entire image may not work well.

1. Global Thresholding

- Uses one threshold for the entire image.
- Simple and fast.
- Not suitable for non-uniform illumination.
- Some object areas may be incorrectly classified.

2. Otsu's Thresholding

- Automatically selects a **single threshold**.
- Works well when the image has clear object and background intensity distributions.
- However, it may not perform well when illumination changes across the image.

3. Adaptive Thresholding

- Uses **different threshold values for different regions** of the image.
- Considers local pixel intensity.
- Works better when lighting is uneven.
- Can separate objects correctly even when different areas have different brightness.

Comparison

Method	Non-Uniform Illumination
Global	Poor
Otsu's	Limited
Adaptive	Best

Suitable Method

Adaptive thresholding is the best choice for non-uniform illumination because it calculates the threshold locally for different regions.

Q15. Explain the Process of Region Growing Segmentation. How Are Seed Points Selected, and How Does the Algorithm Determine Whether Neighboring Pixels Belong to the Same Region?

Definition

Region growing is a region-based segmentation technique that groups pixels having **similar properties** into one region.

Working Process

1. **Select Seed Point**
   One or more starting pixels called **seed points** are selected.
2. **Check Neighboring Pixels**
   The algorithm examines pixels around the seed point.
3. **Compare Similarity**
   The intensity, color, or other property of a neighboring pixel is compared with the region.
4. **Add Similar Pixels**
   If the neighboring pixel satisfies the similarity condition, it is added to the region.
5. **Continue Growing**
   The newly added pixels are used to check their neighboring pixels.
6. **Stop Condition**
   Region growing stops when no more neighboring pixels satisfy the selected similarity condition.

Seed Point Selection

Seed points can be selected:

- Manually by the user.
- Automatically from areas with similar intensity.
- From pixels representing important objects.

Example

Suppose an object contains pixels with intensity values around **100**. If the seed pixel has intensity 100, neighboring pixels with similar values may be added to the same region.

Advantages

- Produces connected regions.
- Useful when objects have similar intensity or color.
- Can provide meaningful object regions.

Q16. Evaluate the Suitability of Edge-Based, Threshold-Based, and Region-Based Segmentation for Medical Image Analysis. Justify Your Choice for Detecting Objects with Poorly Defined Boundaries.

1. Edge-Based Segmentation

- Detects the **boundaries or edges** of objects.
- Useful when object boundaries are clear.
- May not work well when boundaries are weak or unclear.
- Noise can also affect edge detection.

Example: Detecting clear boundaries of bones in an X-ray.

2. Threshold-Based Segmentation

- Separates objects based on **pixel intensity**.
- Simple and fast.
- Works well when the object and background have different intensity values.
- Not suitable when intensity values overlap or illumination is uneven.

Example: Separating bright regions from a dark background.

3. Region-Based Segmentation

- Groups neighboring pixels having **similar properties**.
- Can form complete regions instead of only detecting boundaries.
- More suitable when object boundaries are poorly defined.
- **Region Growing** is a common method.

Comparison

Method	Best Suitable For
Edge-Based	Clear boundaries
Threshold-Based	Different intensity
Region-Based	Similar regions / unclear boundaries

Suitable Choice

For objects with **poorly defined boundaries**, **region-based segmentation** is more suitable because it groups pixels based on their similarity rather than depending only on a strong boundary.

Q17. Explain the Principle of the Hough Transform and Analyze How It Can Be Used for Detecting Straight Lines and Circles in an Image.

Definition

Hough Transform is an image processing technique used to **detect specific shapes** in an image, especially **straight lines and circles**.

Principle

The basic principle is to convert image points into a **parameter space** and find the parameters that receive the highest number of votes.

Line Detection

A straight line can be represented as:

$$\rho = x \cos \theta + y \sin \theta$$

Where:

- ρ = distance from origin
- θ = angle of the line

Steps for Line Detection

1. Detect edges in the image.
2. Consider each edge pixel.
3. Convert the pixel into possible line parameters.
4. Give votes to the corresponding parameter values.
5. The parameters receiving many votes represent a detected line.

Circle Detection

A circle can be represented using:

$$(x - a)^2 + (y - b)^2 = r^2$$

Where:

- **a, b** = center of circle
- **r** = radius

The Hough Transform finds the center and radius by voting in parameter space.

Applications

- Detecting road lanes
- Detecting circular objects
- Detecting shapes in industrial images
- Object recognition

Q18. Explain Erosion, Dilation, Opening, and Closing in Binary Morphological Image Processing. Give One Practical Application of Each Operation.

Morphological image processing is used to analyze and modify the **shape and structure of objects** in binary images.

1. Erosion

Erosion removes pixels from the boundaries of objects.

- Makes objects smaller.
- Removes small unwanted parts.
- Can separate objects that are connected by a thin region.

Application: Removing small noise from a binary image.

2. Dilation

Dilation adds pixels to the boundaries of objects.

- Makes objects larger.
- Fills small gaps.
- Can connect nearby objects.

Application: Connecting broken parts of an object.

3. Opening

Opening = Erosion followed by Dilation

- Removes small objects or noise.
- Preserves the general shape of larger objects.

Application: Removing small noise from scanned documents.

4. Closing

Closing = Dilation followed by Erosion

- Fills small holes and gaps.
- Connects nearby parts of objects.

Application: Filling small gaps in object boundaries.

Simple Difference

- **Erosion → Shrinks objects**
- **Dilation → Expands objects**
- **Opening → Removes small objects/noise**
- **Closing → Fills gaps and holes**

Q19. Explain How Connected Component Analysis, Boundary Extraction, and Hole Filling Can Be Applied to Extract and Analyze Objects in a Binary Image.

These are important techniques used to **identify and analyze objects** in a binary image.

1. Connected Component Analysis

Connected Component Analysis identifies groups of **connected foreground pixels**.

Steps:

1. Scan the binary image.
2. Find connected foreground pixels.
3. Give a unique label to each connected object.
4. Analyze each labeled object separately.

Application: Counting objects in a binary image.

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2. Boundary Extraction

Boundary extraction is used to find the **outer boundary of an object**.

It can be obtained by comparing the original object with its eroded version.

Basic idea:

Boundary = Original Image – Eroded Image

This leaves mainly the boundary pixels.

Application: Finding the shape or outline of an object.

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3. Hole Filling

Hole filling is used to **fill unwanted empty regions inside an object**.

Steps:

1. Identify the background area inside an object.
2. Start from suitable seed points.
3. Grow the background region.
4. Fill the enclosed holes.

Application: Filling holes inside characters or objects.

Example

In a binary image containing letters such as **A, B, and O**, connected component analysis can identify each letter, boundary extraction can find their outlines, and hole filling can fill enclosed regions when required.

Q20. Analyze the Role of Skeletonization and Thinning in Extracting the Structural Characteristics of Objects. Discuss Their Applications in Pattern Recognition, Character Recognition, or Shape Analysis.

Skeletonization

Skeletonization reduces an object to a thin representation called its **skeleton** while preserving its basic structure and shape.

It removes unnecessary pixels from the object while keeping its important structural information.

Thinning

Thinning repeatedly removes boundary pixels from an object until it becomes a **thin line structure**, usually about one pixel wide.

Role in Image Processing

1. **Reduces Data**
    - Removes unnecessary pixels.
    - Makes the object representation smaller.
2. **Preserves Structure**
    - Keeps important shape and connectivity information.
3. **Extracts Shape**
    - Makes it easier to analyze the structure of an object.
4. **Simplifies Recognition**
    - A thin representation is easier for pattern recognition algorithms to process.

Applications

1. Character Recognition

- Used to simplify letters and numbers before recognition.
- Example: Skeleton of handwritten characters.

2. Pattern Recognition

- Helps identify patterns based on their structural features.

3. Shape Analysis

- Used to study the shape and structure of objects.

4. Handwriting Analysis

- Useful for analyzing strokes and connected structures.

Example

A thick handwritten letter can be converted into a **thin skeleton**, making its basic structure easier to recognize and compare.